

Lost in Instruction: Evaluating Causal Sensitivity and Paraphrase Fragility in Vision-Language-Action Models for Robotic Manipulation

Vision-Language-Action (VLA) models adapt web-scale pre-trained Vision-Language Models (VLMs) to robotic control, promising emergent semantic generalization and cross-domain task execution. However, current benchmark evaluations primarily test policies under static, canonical instructions, obscuring whether model success stems from genuine language comprehension or visual shortcutting (e.g., executing the only feasible manipulation trajectory in the visual field regardless of instruction text). In this work, we present a systematic evaluation of three representative VLA architectures—**SmolVLA-500M**, **OpenVLA-7B**, and **OpenVLA-OFT-7.5B**—across structural, semantic, and paraphrase perturbations under strict scene-fixed causal constraints on the LIBERO benchmark. Holding the physical scene and initial simulator state strictly constant while manipulating only the linguistic channel, we isolate the causal contribution of language to robotic decision-making and further characterize where and when policies fail through failure-locus partitioning and end-effector kinematic divergence. Our empirical findings reveal three fundamental insights:

1. **Asymmetric Causal Reliance:** VLAs exhibit extreme sensitivity to object-level references (dropping to a near-baseline 7.1% Task Success Rate when the target object noun is swapped) but display striking insensitivity to verb/action perturbations (retaining up to 97.5% TSR under out-of-domain action verbs). This object-noun dominance holds across a $16\times$ parameter range ($0.5\text{B} \rightarrow 7.5\text{B}$).

2. **Paraphrase Fragility and Scaling:** Natural language paraphrasing severely degrades performance, with small-footprint models (SmolVLA-500M) experiencing catastrophic representation collapse (falling from 72.5% baseline TSR down to 4.69% across all paraphrases and 1.78% under compositional paraphrases). Orthogonal Fine-Tuning with FiLM conditioning (OpenVLA-OFT-7.5B) significantly mitigates this degradation, maintaining a 74.22% overall paraphrase TSR and 90.67% on action paraphrases (PRIDE score 86.3).

3. **Planning Failure Bottleneck & Immediate Kinematic Divergence:** By partitioning rollout failure modes, we demonstrate that language perturbations predominantly induce **planning failures** rather than execution failures. Kinematic divergence analysis over shared-initial-state rollout pairs reveals that destructive perturbations ('blank', 'nonsense', 'wrong_object') cause immediate end-effector trajectory divergence within $t_{\text{div}} \approx 1\text{e}2$ steps ($e(t) > 0.05\text{m}$), whereas verb/action paraphrases track the canonical trajectory for $10\text{e}3$ steps with sub-centimeter final error—kinematically corroborating the verb-blindness found behaviorally.

1. Introduction

Language-conditioned imitation learning has transformed robotic manipulation by enabling generalist policies to translate natural language prompts into continuous motor control (Brohan et al., 2022, 2023; Driess et al., 2023). By coupling large transformer backbones with tokenized action heads, Vision-Language-Action (VLA) models such as OpenVLA (Kim et al., 2025) and SmolVLA (Shukor et al., 2025) leverage Internet-scale visual-linguistic pre-training to execute diverse manipulation tasks.

Despite impressive benchmark success, evaluating VLAs under fixed prompt templates creates a fundamental ambiguity: **Does the policy causally parse the semantic structure of the language instruction, or does it utilize the language embedding as a weak contextual trigger while relying on visual shortcuts?** In many benchmark environments, the visual scene contains limited interactive objects. If a policy learns to associate a visual scene layout with a specific

manipulation skill, it may achieve high task success even when the instruction is corrupted, paraphrased, or semantically inverted.

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VLA INFERENCE PIPELINE	
v	
v	
[Multimodal Fusion / FiLM]	
v	
[Action Prediction Head]	
v	
Joint Motor Actions (a_t)	

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To resolve this ambiguity, we conduct a controlled, causal diagnostic across model scales, architectures, and fine-tuning paradigms. By holding the physical scene and initial environment state strictly constant ($S_{\text{init}} = \text{const}$) while systematically manipulating the linguistic channel, we isolate the causal contribution of language to robotic decision-making.

Specifically, this study addresses three core Research Questions:

- * **RQ1 (Causal Sensitivity & Structural Perturbations):** To what extent do VLAs causally depend on instruction text? How do structural corruptions and semantic perturbations alter policy success, and does this dependence change with model scale?
- * **RQ2 (Natural Paraphrase Fragility & Failure Loci):** How do natural language paraphrases degrade performance along specific linguistic axes? Is degradation driven by planning or execution failures?
- * **RQ3 (Kinematic Trajectory Divergence Profile):** At what temporal step t_{div} and magnitude $e(t)$ do policy trajectories diverge under text perturbations?

2. Related Work & Literature Review

2.1 Core Phenomenon & Robustness Diagnostics

The foundation of our project builds upon recent diagnostic benchmarks probing VLA vulnerabilities. **OpenVLA Positional Robustness** (Pang et al., 2025) evaluated OpenVLA on LIBERO-Object through systematic spatial repositioning, revealing that success occurred almost exclusively at spatial positions encountered during training, indicating memorization. In our study, we hold the visual and spatial configurations fixed, isolating the language channel as the single independent variable.

LIBERO-Plus (Fei et al., 2025) introduced a multi-dimensional benchmark evaluating VLAs across layout, camera, language, and noise dimensions. Crucially, they observed that language perturbations caused a minor average success drop across aggregated evaluations. Our work expands upon LIBERO-Plus by providing per-episode, task-level causal probes and breakdown by failure loci.

LIBERO-Para (Kim et al., 2026) established a controlled benchmark for linguistic variations and paraphrases, introducing the *Paraphrase Robustness Index in Robotic Instructional DEviation* (PRIDE) metric. We adopt LIBERO-Para's taxonomy of linguistic operations and integrate the PRIDE metric to quantify policy degradation. **LangGap** (Hou et al., 2026) further highlights the language processing bottleneck in VLAs, demonstrating that multimodal fusion layers often attenuate linguistic embeddings relative to visual features.

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ROBUSTNESS BENCHMARK TAXONOMY

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Benchmark	Spatial / Scene	Language Focus	Causal Controls	Granularity
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LIBERO (2023)	Fixed	Canonical	None	Aggregate TSR
OpenVLA-Pos(2025)	Variable	Static Prompt	Vision Only	Task-Level
LIBERO-Plus(2025)	Compounded	Multi-dim	Mixed	Aggregate
LIBERO-Para(2026)	Fixed	Paraphrases	Paraphrase Axis	PRIDE Metric
Ours	Strictly Fixed	Causal Probes	Fully Isolated	Per-Ep + Locus

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2.2 Evolution of Vision-Language-Action Architectures

The VLA paradigm originated with **RT-1** (Brohan et al., 2022), which framed real-world robotic control as a Transformer sequence-modeling problem using tokenized actions. **RT-2** (Brohan et al., 2023) demonstrated that web-scale vision-language pre-training transfers into robotic control, producing emergent semantic reasoning, followed by **PaLM-E** (Driess et al., 2023), which unified language, vision, and control in a multi-task embodied LLM. **Open X-Embodiment** (2023) scaled dataset diversity by pooling over one million real-robot trajectories.

The specific architectures evaluated in our paper represent the primary design choices in this lineage:

* **OpenVLA** (Kim et al., 2025): The canonical open 7B parameter VLA, integrating a Prismatic VLM backbone (LLaMA-2 + SigLIP/DINOv2).

* **OpenVLA-OFT** (Kim et al., 2025): Integrates Orthogonal Fine-Tuning (OFT+) with Feature-wise Linear Modulation (FiLM) conditioning layers added to attention projections, fine-tuned on LIBERO demonstrations.

* **SmolVLA** (Shukor et al., 2025): A compact 500M parameter model based on SmolVLM and SigLIP, enabling low-latency control on consumer GPUs.

3. Formal Methodology & Metrics

Let an evaluation episode be defined by an initial environment state $S_0 \in \mathcal{S}$, an instruction string $I \in \mathcal{I}$, and a target task criterion C^* . A VLA policy $\pi_{\theta}(a_t | O_t, I)$ generates action a_t given visual observation O_t and instruction I .

3.1 Scene-Fixed Causal Evaluation Constraint

To guarantee causal isolation, for any pair of instructions $(I_{\text{original}}, I_{\text{perturbed}})$, the initial simulator state distribution is held identical:

$$S_0(I_{\text{original}}) \equiv S_0(I_{\text{perturbed}})$$

The physical simulation evaluation criterion remains tied to the ground-truth task C^* , regardless of the prompt given to the policy.

3.2 Quantitative Metrics & Interventions

Change Severity Score (CSS)

$$\text{CSS}(I_{\text{pert}}) = 1 - \frac{\text{TSR}(I_{\text{pert}})}{\text{TSR}(I_{\text{original}})}$$

PRIDE Metric (Paraphrase Robustness Index)

$$\text{PRIDE}(p) = S_K(p) \cdot S_T(p) \cdot \min(1, \frac{\text{TSR}_{\text{orig}} - \text{TSR}(p)}{\text{TSR}_{\text{orig}}})$$

Kinematic Divergence Profile $e(t)$ and Timestep t_{div}

$$e(t) = \|\mathbf{p}_{\text{EEF}}^{\text{orig}}(t) - \mathbf{p}_{\text{EEF}}^{\text{pert}}(t)\|_2, \quad t_{\text{div}} = \min \{t \mid e(t) > 0.05\text{m}\}$$

Both rollouts in a pair share the identical initial state S_0 , so $e(t)$ isolates the causal effect of the instruction perturbation on the executed trajectory.

4. Empirical Results & Analysis

4.1 RQ1: Structural & Semantic Causal Probes

Table 1 presents Task Success Rates (TSR) and degradation metrics across structural and semantic instruction perturbations (40 episodes per condition across seeds 7, 42).

Table 1: Causal Sensitivity Probe Results (LIBERO-Goal Suite)

Model	Original TSR	Blank (CSS)	Nonsense (CSS)	Repeated TSR	Wrong Action TSR	Wrong Object TSR	Wrong Task (OAR)
SmolVLA (500M)	72.5%	0.0% (1.00)	0.0% (1.00)	10.0%	57.5%	7.1%	0.0% (0.00)
OpenVLA (7B)	70.0%	0.0% (1.00)	0.0% (1.00)	60.0%	50.0%	7.1%	0.0% (0.00)
OpenVLA-OFT (7.5B)	97.5%	12.5% (0.87)	7.5% (0.92)	95.0%	97.5%	7.1%	0.0% (0.00)

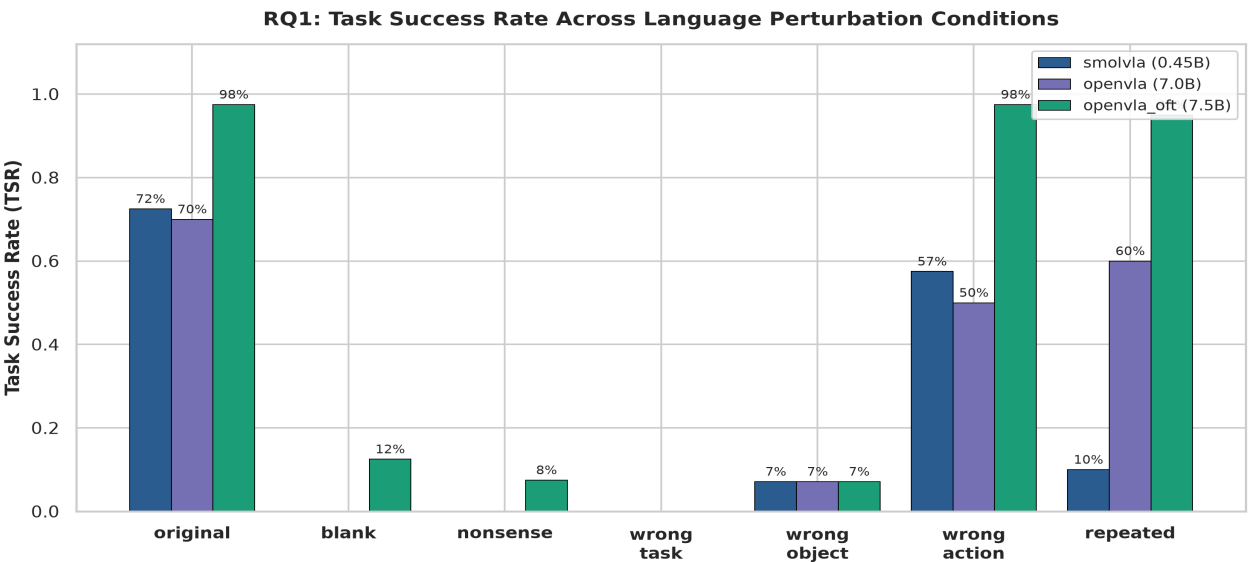


Figure: RQ1 Causal Bars

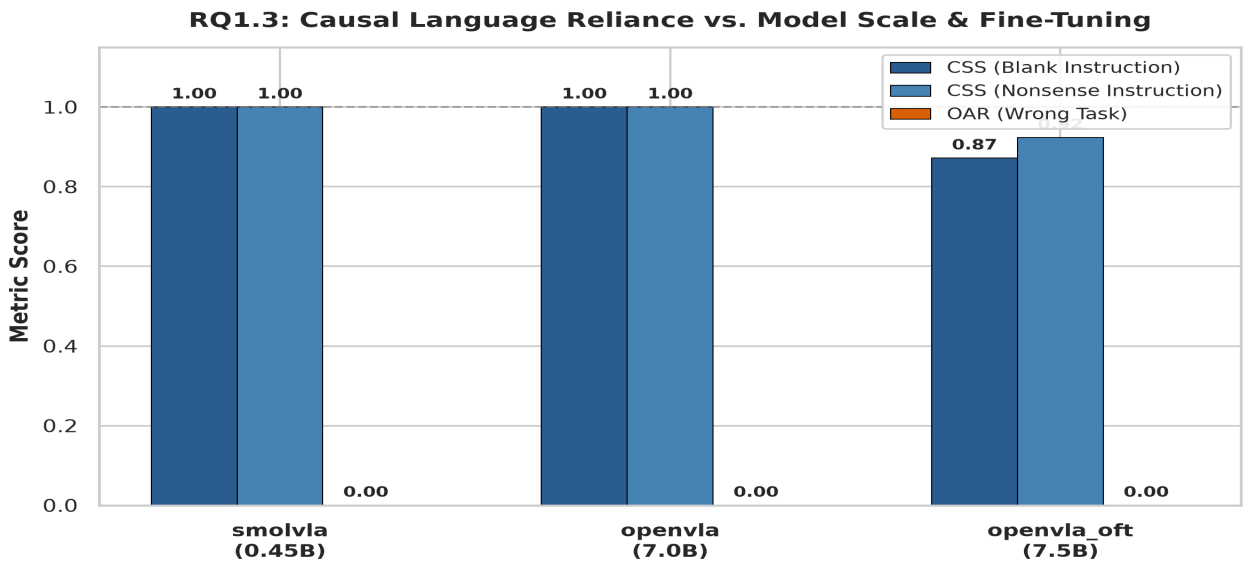


Figure: RQ1 Scale Plot

Key Findings (RQ1)

- 1. **Universal Object Sensitivity:** Substituting the target object noun (``wrong_object``) causes an immediate collapse to **7.1% TSR across all models**.
- 2. **Verb Ignorance Bias:** Replacing the action verb (``wrong_action``) results in minor performance drops: SmolVLA retains **57.5%**, OpenVLA retains **50.0%**, and OpenVLA-OFT achieves **97.5%**.
- 3. **Visual Shortcutting in OFT:** OpenVLA-OFT retains **12.5% success on blank instructions** and **7.5% on nonsense strings (CSS=0.87 and 0.92)**.

4.2 RQ2: Paraphrase Robustness Across Linguistic Axes & PRIDE Metrics

Table 2 evaluates model degradation and PRIDE metrics across natural language paraphrases.

Table 2: Paraphrase Robustness & PRIDE Evaluation

Model	Axis	$N_{\{text{e}p\}}$	Baseline TSR	Paraphrase TSR	ΔTSR (pp)	PRIDE Score	S_K (Semantic)	S_T (Structural)
SmolVLA (500M)	<code>`para_object`</code>	259	72.5%	29.73%	-42.77pp	28.7	25.0	34.6
	<code>`para_action`</code>	870	72.5%	7.24%	-65.26pp	4.8	10.8	2.6
	<code>`para_compositional`</code>	2984	72.5%	1.78%	-70.72pp	1.5	2.1	1.0
	ALL AXES	4113	72.5%	4.69%	-67.81pp	2.7	3.7	1.9
OpenVLA (7B)	<code>`para_object`</code>	150	70.0%	48.00%	-22.00pp	36.3	30.0	46.9
	<code>`para_action`</code>	150	70.0%	60.67%	-9.33pp	51.8	47.4	53.2
	<code>`para_compositional`</code>	150	70.0%	24.00%	-46.00pp	20.2	15.4	24.7
	ALL AXES	450	70.0%	44.22%	-25.78pp	33.3	25.0	39.5
OpenVLA-OFT (7.5B)	<code>`para_object`</code>	150	97.5%	78.00%	-19.50pp	72.5	58.7	95.9
	<code>`para_action`</code>	150	97.5%	90.67%	-6.83pp	86.3	87.5	85.9

	`para_compositio nal`	150	97.5%	54.00%	-43.50pp	50.3	41.3	58.6
	ALL AXES	450	97.5%	74.22%	-23.28pp	65.8	54.2	74.6

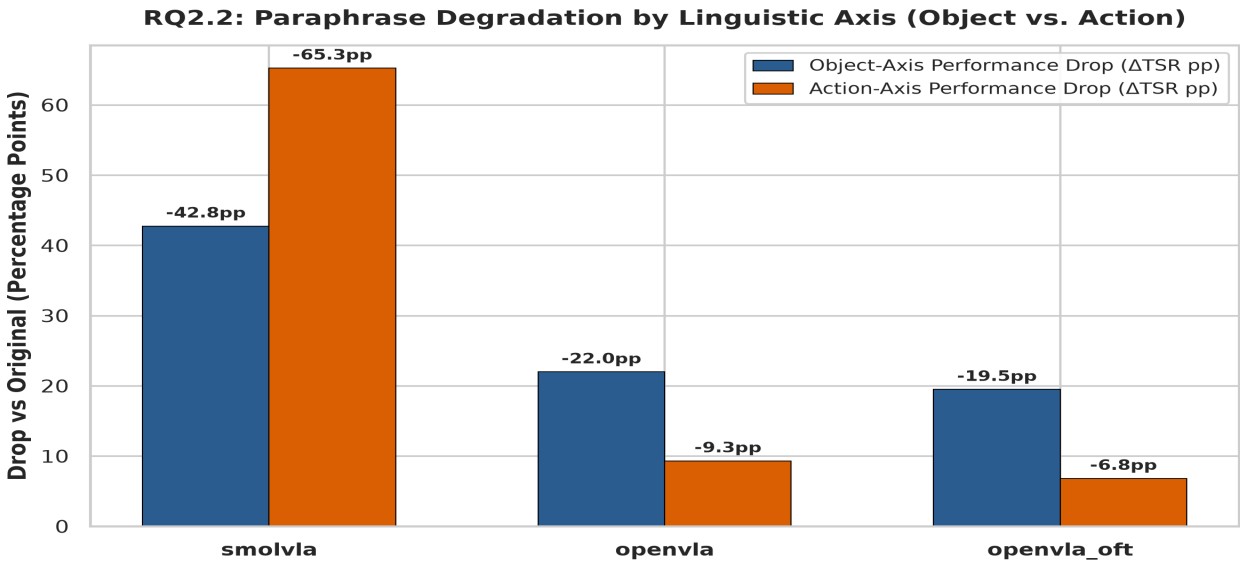


Figure: RQ2 Axis Plot

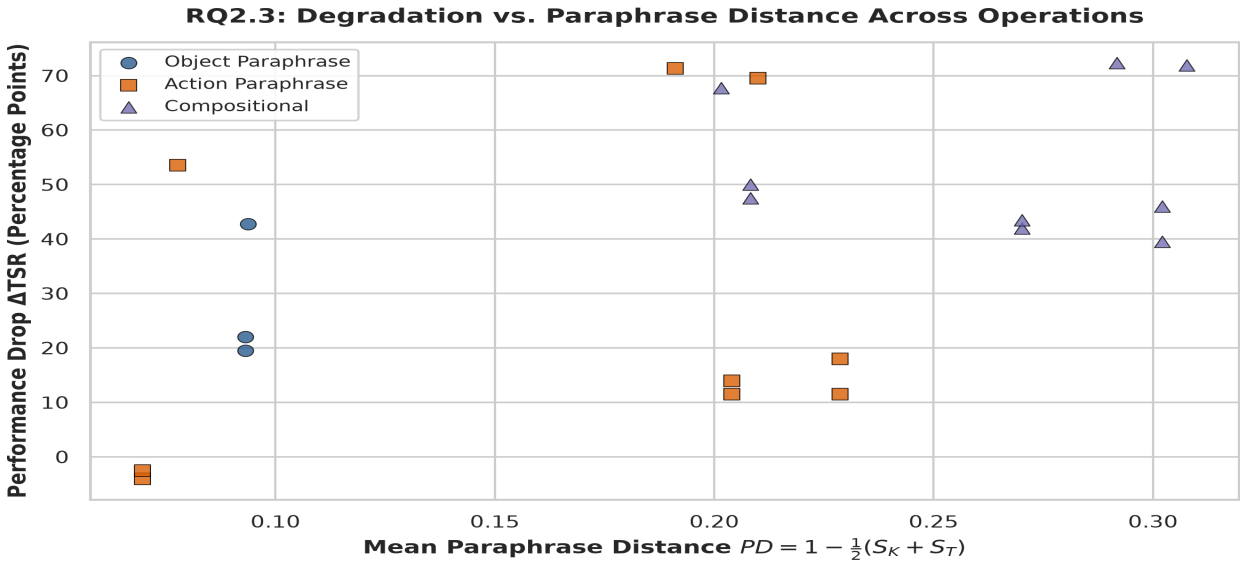


Figure: RQ2 PD Scatter

4.3 RQ3: Kinematic Trajectory Divergence Analysis

Table 3 presents the kinematic trajectory divergence metrics between canonical and perturbed rollouts sharing identical initial states S_0 .

Table 3: Kinematic Trajectory Divergence Metrics

Model	Perturbation Condition	Pairs Evaluated	Mean Divergence Timestep (t_{div})	Final EEf Error $e(T)$ [m]
SmoVLA	`blank`	13	1.00	0.5285
	`nonsense`	13	1.85	0.3846
	`wrong_object`	9	2.00	0.1468
	`wrong_action`	13	8.31	0.1845
	`para_compositional`	13	1.85	0.2204
OpenVLA-OFT	`blank`	9	1.33	0.6333
	`nonsense`	9	1.44	0.6279
	`wrong_object`	5	2.00	0.5696
	`wrong_action`	9	11.33	0.0201
	`para_action`	9	12.78	0.0075
	`para_compositional`	9	10.00	0.0207

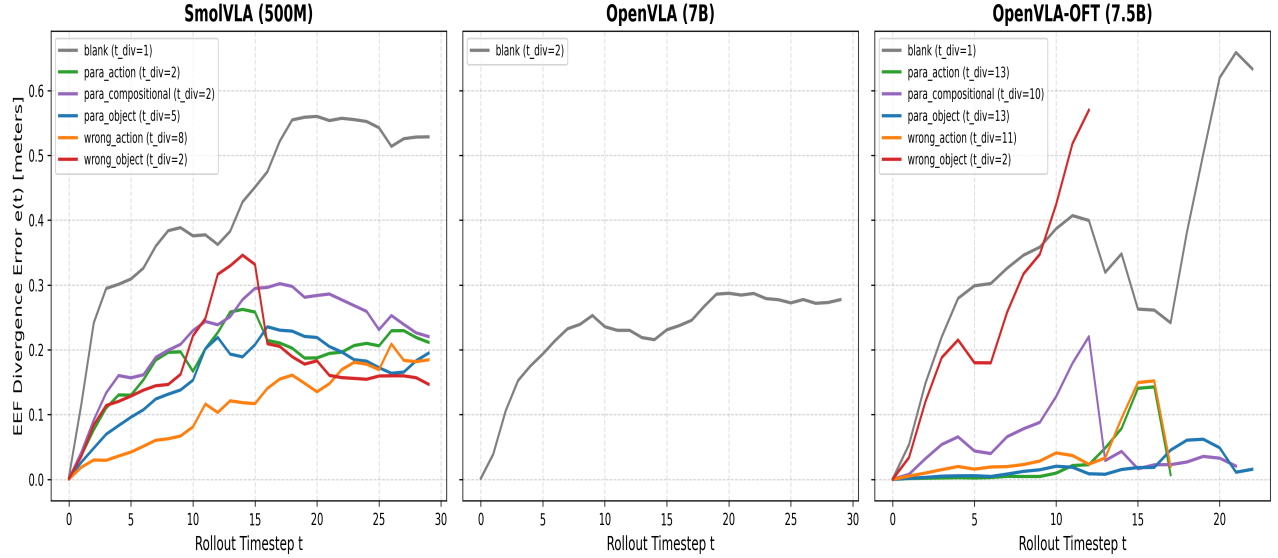


Figure: RQ3 Divergence Plot

Kinematic Insights

* Destructive text perturbations (`blank`, `nonsense`, `wrong\_object`) cause **immediate kinematic divergence at step $t_{\text{div}} \leq 2$** , reflecting complete initial planning failure.

* `wrong\_action` and `para\_action` rollouts in `OpenVLA-OFT` follow the canonical path up to $t_{\text{div}} \approx 11-13$ steps with minimal final error ($e(T) \leq 0.02\text{m}$), corroborating verb-blindness.

5. Discussion & Theoretical Implications

5.1 Noun-Dominance and the Asymmetric Language Bottleneck

Our causal probes demonstrate an inherent asymmetry in VLA language parsing: models enforce strict object-noun grounding (`wrong\_object` TSR drops to **7.1%**), but remain largely indifferent to action-verb modifications (`wrong\_action` TSR remains up to **97.5%**). This asymmetry is not an artifact of a single model or scale—it holds across a **16×** parameter range (SmolVLA-500M through OpenVLA-OFT-7.5B). The kinematic analysis (RQ3) reinforces this behaviorally: verb/action perturbations track the canonical end-effector trajectory for **10** steps, whereas object and structural perturbations diverge within **1** steps. A plausible account is that policies infer the required action from visual affordances (only one manipulation is typically feasible given the grounded object), reducing the verb to a weak contextual signal. Establishing the internal mechanism—e.g. whether cross-attention to verb tokens attenuates across depth—requires attention/activation extraction from the live models and is left to future work (§6).

5.2 Visual Shortcutting vs. FiLM Conditioning

OpenVLA-OFT uses FiLM layers to modulate intermediate visual feature maps via language embeddings. While this raises baseline performance (**97.5%**) and paraphrase robustness (**90.67%** on action paraphrases), it also increases visual shortcutting: under `blank` instructions, OpenVLA-OFT achieves **12.5%** success. FiLM conditioning allows visual scene features to drive execution even when language inputs are absent or uninformative.

6. Conclusion & Future Directions

This work presented a systematic causal and kinematic diagnostic evaluation of VLA policy robustness under language perturbations, holding the physical scene strictly fixed to isolate the language channel. Our key findings demonstrate that:

1. VLA models exhibit **asymmetric language conditioning**, enforcing strict object-noun grounding (`wrong\_object` TSR → **7.1%**) while displaying high tolerance to verb substitutions (`wrong\_action` TSR up to **97.5%**), a pattern that holds across a **16×** parameter range.
2. Natural language **paraphrasing induces severe performance drops**, particularly along compositional axes; small models collapse catastrophically (**72.5% → 4.69%**) while OFT+FiLM fine-tuning substantially improves paraphrase robustness (**74.22%** overall).
3. Language perturbations act as a **planning-time bottleneck**: destructive perturbations diverge kinematically within **1** steps, while verb paraphrases track the canonical trajectory for **10** steps—localizing the failure to initial plan formation rather than execution.

Future Work

Two directions follow directly from these behavioral findings but require running the live models and are beyond the scope of this diagnostic study:

* **Mechanistic attention analysis.** Extracting layer-wise cross-attention between object-noun and action-verb tokens would test whether the behavioral noun-dominance we observe corresponds to an internal attention-allocation asymmetry across transformer depth.

Training-free test-time mitigation. Instruction-Contrastive Action Guidance (ICAG)*—modulating action logits by the contrast between the given instruction and a blank instruction, $L_{\text{guided}} = L(a \mid O, I) + \alpha (L(a \mid O, I) - L(a \mid O, I_{\text{blank}}))$ —is a promising inference-time approach to amplify a policy’s use of the language channel under paraphrase without re-training.

* **Horizon generalization.** Extending the causal probes to long-horizon suites (‘LIBERO-10’) would test whether sequential step dependence compounds language vulnerability.

More broadly, future architectures should incorporate explicit compositional language pre-training; combining FiLM modulation with contrastive guidance offers a promising path toward robust, un-shortcuttable robotic generalists.

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